llmXive Automated Review of Video2GUI: Synthesizing Large-Scale Interaction Trajectories for Generalized GUI Agent Pretraining

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the alignment between claims and cited evidence.

1. Inconsistent Model Citations and Naming In Section 3.2 (“Trajectory Extraction”), the authors state: “we employ **Gemini-3-Pro** directly as the annotation model.” However, the bibliography entry `comanici2025gemini` refers to “**Gemini 2.5**“ (specifically “Gemini 2.5: Pushing the frontier...”). There is no citation for a “Gemini-3-Pro” model. If this is a typo for Gemini 2.5, the claim is supported by the citation but the text is factually incorrect regarding the model version. If “Gemini-3-Pro” is a distinct, unreleased, or internal model, the claim is unsupported by the provided bibliography, and the reproducibility of the “high-quality” annotation claim is compromised. This discrepancy appears in the main text and the Appendix (Section “Action Spatial Grounding Details”).

2. Overstated Performance Improvements The Abstract and Introduction claim that pre-training yields “consistent and substantial improvements of **5–20%**“ across benchmarks.

- **Support:** Table 1 shows a 15.1% gain for Qwen2.5-VL on ScreenSpot-Pro and a 12.9% gain for Mimo-VL on OSWorld-G.
- **Contradiction:** Table 2 shows the improvement for Qwen2.5-VL on **AndroidControl-Low (Type Acc)** is only **0.8%** (94.1% to 94.9%).

The claim of a “5–20%” range is not supported by the full set of results presented in the paper, as at least one metric falls significantly below the stated lower bound. The text should be qualified to reflect the variance in gains or the specific benchmarks where this range holds.

3. Unsubstantiated Manual Verification Metrics In Section 3.3 (“Action Spatial Grounding”), the authors claim: “Manual verification of 200 randomly sampled actions confirms that over **95%** are accurately parameterized.”

- **Missing Evidence:** The text does not define the ground truth criteria used for this verification (e.g., Intersection over Union threshold, exact coordinate tolerance).
- **Missing Methodology:** It does not state who performed the verification (e.g., number of annotators, expertise level) or the inter-annotator agreement (Krippendorff’s alpha) for this specific 200-sample check. While Section 5.2 mentions a user study with 300 samples and an alpha of 0.84, the specific 95% accuracy claim for the 200-sample grounding check lacks the necessary methodological detail to be considered a verified fact.

4. Conflation of Model Results in Scaling Analysis In Section 5.1 (“Scaling Effects”), the text states: “On ScreenSpot-Pro... the model starts at approximately 41% accuracy and reaches a peak of **56.9%** at 200 billion tokens.”

- **Discrepancy:** According to Table 1, the **Qwen2.5-VL-7B** model (the one typically associated with the "41%" baseline in the text's narrative flow) reaches **41.9%** after training. The **56.9%** score belongs to the **Mimo-VL-7B** model. The text appears to conflate the performance of the two different base models, attributing the higher score to the scaling curve of the model that actually achieved the lower score. This misrepresents the scaling law results for the specific model being discussed.

Action items.

- [**writing**] In Section 3.2 (Trajectory Extraction), the paper claims to use 'Gemini-3-Pro' for annotation. The bibliography lists 'Gemini 2.5' (comanici2025gemini). Verify if 'Gemini-3-Pro' is a typo or a distinct model not cited, as this affects the reproducibility of the annotation quality claims.
- [**writing**] The abstract and Section 1 claim pre-training yields 'consistent and substantial improvements of 5–20%'. Table 1 (grounding benchmark) shows a 15.1% gain for Qwen2.5-VL on ScreenSpot-Pro Avg, but Table 2 (offline benchmark) shows a 0.8% gain for Qwen2.5-VL on AndroidControl-Low Type Acc. The '5-20%' range is not supported by the full dataset of results presented.
- [**science**] Section 3.3 claims 'Manual verification of 200 randomly sampled actions confirms that over 95% are accurately parameterized'. The text does not define the criteria for 'accurately parameterized' (e.g., IoU threshold, exact coordinate match) or the inter-annotator agreement for this manual check, making the 95% figure unverifiable.
- [**writing**] Section 5.1 (Scaling Effects) states the model reaches '56.9% at 200 billion tokens' on ScreenSpot-Pro. However, Table 1 lists the final score for Mimo-VL-7B + WildGUI as 56.9, while the text implies this is the Qwen2.5-VL result (which is 41.9 in the table). The text conflates the results of the two different base models.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure displays a raw sequence of GUI interactions and data logs but is rendered as a disjointed composite rather than a polished scientific illustration. The caption is a non-descriptive placeholder that fails to explain the figure's content or relevance to the paper's claims.

Figure 2

The figure clearly illustrates a 5-step GUI interaction workflow with screenshots and text, but the caption is a generic placeholder that does not describe the content or the specific task shown.

Figure 3

The figure presents a detailed step-by-step interaction log but lacks a descriptive caption and cohesive visual design, rendering it difficult to interpret as a standalone scientific illustration.

Figure 4

The figure effectively illustrates a multi-step GUI interaction but is critically undermined by a non-descriptive placeholder caption that fails to explain its content or purpose.

Action items.

- [**fatal**] Figure 1: The caption 'example. [ZvNgczioehg_task_0.png]' is a placeholder that fails to describe the figure's content (a 5-step GUI interaction workflow), making it impossible to understand the figure's purpose or claims without reading the internal text.
- [**science**] Figure 1: The figure is a raw composite of screenshots and JSON data logs rather than a synthesized scientific visualization; it lacks a cohesive layout, clear visual hierarchy, or annotations that explain the significance of the steps to the reader.
- [**writing**] Figure 2: The caption 'example. [EF-Qh1ydNgk_task_4.png]' is a placeholder that fails to describe the figure's content (a 5-step GUI interaction workflow), making it impossible to understand the figure's purpose without reading the internal text.
- [**writing**] Figure 2: The internal text 'Instruction: Set the computer's power plan to High Performance' is not referenced in the caption, creating a disconnect between the figure label and the visual data.
- [**writing**] Figure 3: The caption 'example. [55BzMmeagwU_task_0.png]' is generic and fails to describe the figure's actual content (a step-by-step iPhone Dark Mode tutorial), making it impossible to understand the figure's purpose without reading the internal text.
- [**science**] Figure 3: The figure is a composite of screenshots and raw JSON-like logs without a unifying visual layout or explanatory annotations, making it difficult to distinguish between the system's 'thought' process and the actual GUI state changes.
- [**writing**] Figure 4: The caption 'example. [oXoXz11H4Q4_task_0.png]' is a placeholder and does not describe the figure's content (a step-by-step guide to factory resetting a Tecno smartphone), making it impossible to understand the figure's purpose or context.
- [**science**] Figure 4: The figure displays a sequence of 7 steps for a smartphone task, but lacks a clear title or introductory text explaining the overall objective, which is only partially inferred from the text within the figure itself.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific jargon that obscures meaning for non-specialist readers. The most critical issue is the unexplained use of acronyms. "VLM" (vision-language model) appears in Section 3 and Section 5 without being defined at first use. Similarly, "POMDP" is introduced in Section 2 but the acronym is used exclusively thereafter, assuming the reader is already familiar with reinforcement learning terminology.

The term "grounding" is used extensively (e.g., "action spatial grounding," "GUI grounding") without a plain-English explanation. While standard in the field, a general reader may not understand that this refers to "mapping text instructions to specific screen coordinates." The phrase "coarse-to-fine" is also used as a technical label for the filtering process; replacing this with "broad-to-specific" or "initial screening followed by detailed evaluation" would improve accessibility.

Additionally, the terms "Stage 1" and "Stage 2" are used frequently to describe the training pipeline (Section 4, 5) but are not explicitly defined in the main text as "Continual Pre-training" and "Post-training" until later or in the appendix. This creates a barrier to understanding the methodology for readers not deeply embedded in the specific training paradigms of this sub-field. Finally, "trajectory" is used as a noun for a sequence of actions without a brief definition, which might be unclear to those unfamiliar with agent-based terminology.

Action items.

- [**writing**] Define 'VLM' at first use. The acronym appears in Section 3 ('VLM-driven approach') and Section 5 ('VLM-based GUI agents') without prior expansion. Use 'vision-language model (VLM)' on first occurrence.
- [**writing**] Replace 'grounding' with 'locating' or 'identifying' in non-technical contexts. The term 'spatial grounding' is used repeatedly (e.g., Section 3.3, 4.2) without a plain-English definition for non-specialists. Consider 'mapping actions to screen coordinates' for clarity.
- [**writing**] Define 'POMDP' at first use. Section 2 introduces 'Partially Observable Markov Decision Process (POMDP)' but the acronym is then used exclusively. Ensure the full term is clear to readers outside reinforcement learning.
- [**writing**] Clarify 'Stage 1' and 'Stage 2' terminology. These terms are used frequently (e.g., Section 4, 5) without a clear, consistent definition of what they refer to (pre-training vs. post-training) in the main text before the appendix.
- [**writing**] Replace 'coarse-to-fine' with 'broad-to-specific' or 'initial screening followed by detailed evaluation' in Section 3.1. While common in CS, the phrase is jargon-heavy for a general audience.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent logical flow from the problem definition (data scarcity) to the proposed solution (Video2GUI) and its validation. The causal chain linking the coarse-to-fine filtering strategy to the resulting dataset quality is well-supported by the human evaluation metrics in Section 5.3. The ablation studies in Section 5.4 logically demonstrate the necessity of the specific loss components, as removing $\mathcal{L}_{traj}$ or $\mathcal{L}_{ground}$ leads to predictable performance drops in planning and grounding tasks respectively.

However, there are minor inconsistencies in the reporting of numerical results that affect the internal consistency of the text versus the tables. Specifically, in Section 5.1, the text reports a ScreenSpot-Pro score of 41.2 for the Qwen2.5-VL-7B variant, whereas Table 2 clearly lists 41.9. Similarly, the scaling analysis in Section 5.2 conflates the specific model results when describing the peak performance on the scaling curve, making it slightly ambiguous which model's trajectory is being described at the 200B token mark. Additionally, the methodology relies on "Gemini-3-Pro," but the bibliography only provides a reference for "Gemini 2.5," creating a gap between the cited evidence and the claimed tool. These issues are fixable through careful editing and reference updates.

Action items.

- [**writing**] In Section 5.1 (Main Results), the text claims Qwen2.5-VL-7B + WildGUI improves from 26.8 to 41.2 on ScreenSpot-Pro. However, Table 2 explicitly lists the result as 41.9. This numerical inconsistency undermines the precision of the reported gains.
- [**writing**] In Section 5.2 (Scaling Effects), the text states the model reaches a peak of 56.9% on ScreenSpot-Pro at 200B tokens. Table 2 reports the final score for Mimo-VL-7B + WildGUI as 56.9, but the text implies this is the Qwen2.5 result or conflates the two models. The scaling curve description needs to explicitly map the specific model architecture to the reported peak value to avoid ambiguity.
- [**writing**] In Section 3.2 (Trajectory Extraction), the authors claim to use Gemini-3-Pro for annotation. However, the bibliography (example_paper.bib) contains a citation for 'Gemini 2.5' (comanici2025gemini) but no entry for 'Gemini-3-Pro'. The reference list must be updated to include the specific source for the model version cited in the methodology.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the scale, quality, and performance impact of the proposed WildGUI dataset and Video2GUI framework. While the methodology is generally sound, there are instances where the authors extrapolate beyond the specific evidence provided in the text, particularly regarding specific numerical metrics and the definition of "state-of-the-art."

First, the claim in the Introduction that "Human evaluation shows that the overall annotation accuracy exceeds 80%" is a significant over-claim. The "Data Quality Check" section (Section 5.4) details a user study where experts rated data on a 1-5 Likert scale, achieving a mean score of 4.62. It does not report a binary accuracy metric (e.g., percentage of trajectories perfectly matching a ground truth). Conflating a high Likert score with a specific "80% accuracy" figure is not supported by the data presented and misrepresents the nature of the evaluation. This specific number should be removed or replaced with the actual metric reported (the Likert score).

Second, the claim that the dataset spans "over 1,500 applications and websites" (Abstract, Table 1) is presented as a definitive statistic. However, the paper does not describe the methodology used to count these unique entities. It is unclear if this count is derived from a unique string match of app names, a clustering of domains, or a manual verification process. Given the potential for noise in metadata (e.g., "Chrome" vs. "Google Chrome"), presenting this as a precise count without a definition of the counting protocol risks over-claiming the diversity of the dataset.

Third, the assertion that the method "surpasses state-of-the-art performance" (Abstract) requires nuance. In Table 2, the proposed model (Mimo-VL-7B + WildGUI) achieves 67.6 on OSWorld-G, surpassing Seed1.5-VL (62.9). However, Seed1.5-VL is a proprietary model, and the paper does not provide details on its training compute, data scale, or model size relative to the proposed 7B model. Claiming to surpass SOTA without qualifying that the comparison includes proprietary models with potentially different resource constraints can be seen as an over-extrapolation of the results' generalizability.

Finally, the claim of processing "500 million video metadata entries" is impressive but lacks temporal context. Without specifying the time window of the data collection, the claim of "generalization" to current real-world interfaces is slightly weakened, as GUI layouts and applications change rapidly. While this is a minor point, it contributes to the overall impression of over-reaching on the dataset's current relevance.

The authors should revise the text to align specific numerical claims (like the 80% accuracy) with the actual metrics reported in the experiments and clarify the definitions used for dataset statistics.

Action items.

- [**writing**] The claim that the dataset spans 'over 1,500 applications and websites' (Abstract, Table 1) lacks a specific methodology for counting unique entities. Without a definition of how 'application' is distinguished from 'website' or how duplicates are handled, this specific number appears to be an over-estimate or an unverified extrapolation from the raw video count.

- [**science**] The statement that 'Human evaluation shows that the overall annotation accuracy exceeds 80%' (Introduction, paragraph 4) is unsupported by the provided text. The 'Data Quality Check' section (Section 5.4) only reports a Likert-scale user study (score 4.62/5) and inter-rater agreement, not a binary accuracy metric against a ground truth. This specific percentage is an over-claim not backed by the reported evidence.
- [**writing**] The claim that the model 'surpasses state-of-the-art performance' (Abstract) is potentially misleading. Table 2 shows the proposed model (Mimo-VL-7B + WildGUI) scoring 67.6 on OSWorld-G, while Seed1.5-VL scores 62.9. However, Seed1.5-VL is a proprietary model, and the paper does not clarify if the comparison is fair regarding model size, training compute, or access to private data. The phrasing implies a universal SOTA without qualifying the proprietary nature of the competitor.
- [**writing**] The paper claims to process '500 million video metadata entries' (Abstract) but does not explicitly state the source or the time window of this data. Given the rapid evolution of YouTube content, the lack of temporal context for this massive dataset makes the claim of 'generalization' slightly over-reaching without specifying the data freshness.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a large-scale framework for synthesizing GUI interaction data from internet videos. While the technical contribution is significant, the paper currently lacks sufficient depth regarding safety, ethics, and legal compliance, which are critical for datasets intended to train autonomous agents.

Dual-Use and Societal Impact: The current "Impact Statement" (Section Impact Statement) is a boilerplate declaration that does not address the specific risks associated with training agents to autonomously control user interfaces. GUI agents possess significant dual-use potential; they could be weaponized for automated credential theft, bypassing CAPTCHAs, executing unauthorized financial transactions, or automating spam and fraud campaigns. The authors must expand this section to explicitly discuss these risks, potential misuse scenarios, and the specific safeguards or limitations they propose (e.g., watermarking synthetic data, restricting release of certain high-risk subsets, or providing usage guidelines).

Data Privacy and Copyright: The dataset "WildGUI" is constructed from 500 million YouTube videos. The paper does not address the legal and ethical implications of scraping and redistributing this content.

1. **Terms of Service:** The authors must confirm that their scraping and processing pipeline complies with YouTube's Terms of Service and the robots.txt policies of the source platforms.

2. **Copyright:** The manuscript should clarify the copyright status of the source videos and the legal basis for creating a derivative dataset (WildGUI). Are the videos used under fair use, or are they licensed? If the dataset is released, does it include copyrighted material?
3. **PII:** The pipeline extracts screenshots and trajectories. The authors should describe any automated or manual processes used to detect and redact Personally Identifiable Information (PII), such as user names, email addresses, or sensitive documents visible in the screenshots, to prevent privacy violations.

Human Subjects Research: Section "Data Quality Check" describes a user study involving five expert participants. The manuscript does not state whether this study received Institutional Review Board (IRB) approval or if informed consent was obtained from the participants. Given that the study involves human evaluation of data, standard ethical protocols for human subjects research should be followed and documented.

Recommendation: The paper requires a minor revision to address these ethical and safety gaps. The authors should add a dedicated subsection in the "Impact Statement" or "Ethical Considerations" to discuss dual-use risks, legal compliance regarding data scraping, PII handling, and IRB approval for the human study.

Action items.

- [**writing**] The Impact Statement (Section Impact Statement) is generic and fails to address specific dual-use risks of autonomous GUI agents, such as unauthorized automation, credential theft, or bypassing security controls. A detailed discussion of potential misuse and mitigation strategies is required.
- [**writing**] The data collection pipeline relies on scraping 500 million YouTube videos. The manuscript lacks explicit confirmation of compliance with YouTube's Terms of Service, copyright laws, and the specific licenses of the source videos. Clarification on legal basis for data usage and redistribution is needed.
- [**writing**] The human evaluation study (Section Data Quality Check) involves five expert participants but does not mention IRB approval, informed consent procedures, or how participant anonymity and data privacy were protected during the study.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a large-scale data synthesis pipeline, but the scientific evidence supporting the quality and efficacy of the resulting dataset relies heavily on point estimates without sufficient statistical rigor.

First, the human evaluation of data quality (Section 5.4) is based on a sample of only 300 data points rated by five experts. While the inter-rater agreement (Krippendorff's α = 0.84) is reported, the sample size is insufficient to generalize the "superior quality" claim to a corpus of 12 million trajectories. The confidence intervals for the mean scores (e.g., 4.62 vs 4.05) are not provided, making it difficult to assess if the difference is statistically significant or driven by sampling variance.

Second, the ablation studies (Table 4) and scaling law analysis (Figure 4) present results from single experimental runs. In deep learning, particularly with large models and stochastic training, performance metrics can vary significantly between seeds. The claim that removing $\mathcal{L}_{traj}$ causes a "significant drop" or that performance scales "consistently" lacks the necessary error bars or standard deviations. Without multiple seeds, it is impossible to rule out that the observed improvements are artifacts of specific initialization or data shuffling.

Third, the validation of the automated pipeline components relies on small sample sizes. The claim of >95% accuracy for the spatial grounding stage is derived from only 200 manually verified samples (Section 3.3). For a binary accuracy estimate of 0.95, a sample of 200 yields a 95% confidence interval of approximately [0.91, 0.98]. While this suggests high accuracy, the width of the interval is non-trivial for a pipeline intended to generate millions of samples. A larger validation set or a more rigorous statistical test is required to substantiate the reliability of the automated annotation.

Finally, the online evaluation results (Figure 2) show performance gains (e.g., 16.4% to 31.9% on AndroidWorld) but do not report the variance across multiple evaluation runs or seeds. Online benchmarks often have high variance due to environmental stochasticity. The absence of error bars makes the magnitude of the improvement difficult to interpret scientifically.

To strengthen the scientific evidence, the authors should report standard deviations or confidence intervals for all quantitative results, increase the sample sizes for human evaluations and automated pipeline validation, and ideally, repeat key experiments (ablation and scaling) with multiple random seeds.

Action items.

- [**science**] The human evaluation study (Section 5.4) relies on a sample size of only 300 data points rated by 5 experts. For a dataset of 12M trajectories, this sample is statistically underpowered to support the claim of 'superior quality' across the entire corpus. Please report confidence intervals or conduct a power analysis to justify the sample size, or increase the sample size significantly.
- [**science**] The ablation study (Table 4) reports performance drops (e.g., AndroidWorld 31.9 -> 24.1) without providing standard deviations or results from multiple random seeds. Given the stochastic nature of LLM training and evaluation, single-run results are insufficient to claim the 'necessity' of specific loss components. Please provide variance metrics or re-run experiments with multiple seeds.
- [**science**] The scaling law analysis (Figure 4) claims a 'strong positive correlation' and 'no saturation' based on a single curve. Without error bars or confidence intervals on the performance metrics at each data scale point (0, 50B, 200B tokens), it is impossible to determine if the observed gains are statistically significant or within the noise of the

evaluation protocol.

- [**science**] The paper claims 95% accuracy for the spatial grounding stage based on 'manual verification of 200 randomly sampled actions' (Section 3.3). This sample size is too small to robustly estimate a 95% accuracy rate with a narrow confidence interval. Please provide the 95% confidence interval for this accuracy estimate or increase the validation sample size.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the evaluation and quality assurance sections requires strengthening to support the strong claims made regarding the dataset's efficacy and quality.

First, in **Section 5.3 (Data Quality Check)**, the authors report a Krippendorff's α of 0.84 based on 300 samples rated by five experts. While this indicates strong agreement, the sample size is negligible compared to the 12 million trajectory dataset. The review requires a **95% confidence interval** for the α coefficient to assess the precision of this estimate. Furthermore, the text does not specify if the 300 samples were randomly drawn or stratified across the 1,500+ applications and platforms. If not stratified, the reported quality score may suffer from selection bias, over-representing common or easy-to-annotate tasks.

Second, the **Scaling Effects analysis (Section 5.1, Figure 4)** presents performance curves as single point estimates. In deep learning experiments, performance variance across random seeds or training runs is non-trivial. The absence of **error bars, standard deviations, or confidence intervals** makes it difficult to determine if the observed "strong positive correlation" is statistically robust or if the gains at specific token counts (e.g., 50B vs 100B) are within the margin of error. The authors should report results averaged over multiple seeds or provide confidence intervals for the accuracy metrics.

Finally, the **Ablation Studies (Section 5.2, Table 4)** report specific performance drops (e.g., AndroidWorld dropping from 31.9% to 24.1% when removing $\mathcal{L}_{traj}$). Without reporting the **variance of these metrics** or conducting statistical significance tests (e.g., paired t-tests or bootstrap resampling), the claim that these drops are "significant" remains anecdotal. The evaluation protocol for online benchmarks (OSWorld, AndroidWorld) often involves stochastic environments; thus, establishing statistical significance is crucial to validate the necessity of specific loss components.

The paper currently lacks the statistical depth required to distinguish between genuine methodological improvements and random variance in the reported benchmarks.

Action items.

- [**science**] Section 5.3 (Data Quality Check): The reported Krippendorff's alpha of 0.84 is strong, but the sample size (300 points rated by 5 experts) is small relative to the 12M dataset. Please report the 95% confidence interval for the alpha coefficient and clarify if the 300 samples were stratified by platform/application to ensure representativeness.

- [**science**] Section 5.1 (Scaling Effects): The scaling law analysis (Figure 4) presents point estimates without error bars or confidence intervals. Given the stochastic nature of training, please report the standard deviation or 95% CI across multiple seeds (or at least clarify if results are from a single run) to validate the statistical significance of the observed gains.
- [**science**] Section 5.2 (Ablation Studies): The ablation results in Table 4 show performance drops (e.g., AndroidWorld 31.9 -> 24.1). Without reported variance or statistical significance tests (e.g., paired t-tests or bootstrap CIs), it is unclear if these drops are statistically significant or within the noise margin of the evaluation protocol.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with a clear logical flow and well-structured arguments. The abstract effectively summarizes the contributions, and the introduction successfully motivates the research problem. The transition between the problem formulation, the proposed Video2GUI framework, and the experimental results is smooth and easy to follow.

However, there are a few specific instances of typographical errors and minor phrasing issues that require correction before publication. Most notably, in Section 5.2 ("Scaling Effects"), the sentence "accuracy improves improves from approximately 55%..." contains a repeated verb. Additionally, in Section 5.1 ("Main Results"), the text "surpassing both Qwen3-VL-32B 60.6and the Seed1.5-VL 45.2" contains Chinese characters (" ") and missing spaces, likely due to a copy-paste error or incomplete translation. These must be corrected to ensure professional polish.

The use of technical terminology is generally precise, though some sentences in the methodology section (e.g., Section 3.1) could be slightly tightened for conciseness. For instance, the description of the metadata filtering process is slightly repetitive regarding the "coarse" nature of the step. Overall, the writing quality is strong, and these issues are easily fixable with a careful proofread.

Action items.

- [**writing**] In Section 5.2 (Scaling Effects), the phrase 'accuracy improves improves' contains a repeated verb. Please correct to 'accuracy improves'.
- [**writing**] In Section 5.1 (Main Results), the sentence 'surpassing both Qwen3-VL-32B at 60.6 and Seed1.5-VL at 62.9' contains a Chinese character ' ' (de) instead of 'at' or 'with'. Please replace with English prepositions.

- **[writing]** In Section 3.1 (Meta Info Filtering), the phrase 'DeepSeek-V3 demonstrates high alignment with human judgment' is vague. Consider specifying the metric (e.g., 'achieves 92% agreement') or citing the specific validation result mentioned in the same paragraph.
- **[writing]** In Section 4.1 (Implementation Details), the phrase '160 CPU cores, 512 GB system memory, and 256 NVIDIA GPUs' lacks specific GPU model names (e.g., A100, H100). While not strictly a writing error, adding the model name improves clarity and reproducibility of the hardware description.